Emilie Yu

CNRS LIRIS lab, Lyon, France

Curriculum Vitae

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Education

Inria, Université Côte d'Azur *2020 - 2023*

PhD in Computer Science – *supervised by Adrien Bousseau*

Technical University of Denmark *2018 - 2020*

MSc in Digital Media Engineering – *Master thesis supervised by J. Andreas Bærentzen*

CentraleSupélec *2015 - 2017*

BSc and MSc in Engineering – *3rd top Engineering School in France*

Research Experience

CNRS – LIRIS *Oct. 2026 - now*

Postdoctoral researcher – *supervised by Aurélien Tabard*

UCSB – Expressive Computation Lab *May 2024 - Jul. 2026*

Postdoctoral researcher – *supervised by Jennifer Jacobs*

University of Toronto – Dynamic Graphics Project *Summer 2023*

Visiting Researcher – *supervised by Fanny Chevalier and Karan Singh*

Adobe Research *Summer 2022*

Research Intern – *supervised by Rubaiat Habib Kazi and Wilmot Li*

Inria, Université Côte d'Azur *Feb. 2020 - Aug. 2020*

Research Intern – *supervised by Adrien Bousseau*

Publications

Machine In The Stack: CNC Machine Motion Programming at Execution Time
Emilie Yu, Benjamin F Ancho, Taneesha Panda, Jennifer Jacobs
In Proceedings of the ACM Symposium on Computational Fabrication, SCF '26
paper </> code

StepDance: A Toolkit for Redesigning CNC Machines Using Physical Metaphors
Ilan E Moyer, **Emilie Yu**, Devon Frost, Maria Yang, Jennifer Jacobs
In Proceedings of the ACM Conference on Human Factors in Computing Systems, CHI '26
webpage paper </> code

Clay ARtools: Precise Machine Toolpath Editing for Clay 3D Printing With Craft-Inspired Direct Manipulation Tools in AR
Joyce E Passananti, **Emilie Yu**, Timea Tihanyi, Tobias Höllerer, Jennifer Jacobs
In Proceedings of the ACM Conference on Human Factors in Computing Systems, CHI '26
webpage paper

WORM: Programming Collaborative Robots Through Manual Actions for Craft-Aligned Digital Fabrication
Samuelle Bourgault, Alejandro Aponte, Megumi Ondo, **Emilie Yu**, Jennifer Jacobs
In Proceedings of the ACM Symposium on User Interface Software and Technology, UIST '25
webpage paper

Towards a sustainable use of GPUs in Graphics Research
Emilie Yu*, Élie Michel*, Octave Crespel, Axel Paris, Felix Hähnlein (* equal contribution)
SIGGRAPH Talk 2025
webpage paper </> code

texTile: Making and Re-making Crochet Granny Square Garments Through Computational Design and 3D-printed Connectors
Ashley Del Valle, Jennifer Jacobs, **Emilie Yu**
In Proceedings of the ACM Conference on Designing Interactive Systems (DIS), 2025
webpage paper

3D-Layers: Bringing Layer-Based Color Editing to VR Painting
Emilie Yu, Fanny Chevalier, Karan Singh and Adrien Bousseau
ACM Transactions on Graphics, presented at SIGGRAPH 2024
webpage paper code

VideoDoodles: Hand-Drawn Animations on Videos with Scene-Aware Canvases
Emilie Yu, Kevin Blackburn-Matzen, Cuong Nguyen, Oliver Wang, Rubaiat Habib Kazi and Adrien Bousseau
ACM Transactions on Graphics, presented at SIGGRAPH 2023
webpage paper code

Piecewise-Smooth Surface Fitting onto Unstructured 3D Sketches
Emilie Yu, Rahul Arora, J. Andreas Bærentzen, Karan Singh, and Adrien Bousseau
ACM Transactions on Graphics, presented at SIGGRAPH 2022
webpage paper code

CASSIE: Curve And Surface Sketching in Immersive Environments
Emilie Yu, Rahul Arora, Tibor Stanko, J. Andreas Bærentzen, Karan Singh, and Adrien Bousseau
In Proceedings of the ACM Conference on Human Factors in Computing Systems, CHI '21
webpage paper code

Professional Experience

Unity Technologies *2019 - 2020*
Continuous integration, continuous deployment – *Infrastructure support and software development*

Datawords *Jul. 2017 - Dec. 2017*
Fullstack web developer

Août Secours Alimentaire (food aid association) *Summer 2016*
Logistics assistant manager

Awards

PhD Dissertation Award – french Informatics Society (SIF) *2024*
Awarded based on the thesis manuscript and jury report.

PhD Dissertation Award – french Computer Graphics Association (GdR IG-RV) *2024*
Awarded based on the thesis manuscript and jury report.

Best Demo Award at IHM 2024 *2024*
VideoDoodles: Hand-Drawn Animations on Videos with Scene-Aware Canvases
Francophone human-computer-interaction conference (IHM 2024)
Best out of 45 demos

Mitacs Globalink Research Award *2023*
Awarded for a 3-months research collaboration at the University of Toronto – *6,000 CAD*

WiGRAPH Rising Stars in Computer Graphics *2022 - 2023*
Academic Career Workshop in Siggraph – *Travel funding for attending Siggraph twice*

Recipient of Best Paper Honorable Mention Award *2021*
CASSIE: Curve And Surface Sketching in Immersive Environments
ACM Conference on Human Factors in Computing Systems (CHI).
Top 5% of submissions

Teaching

Creative Motion Control – Media Arts and Technology MSc, UCSB *Spring 2026*
Co-lecturer

Digital Fabrication – Media Arts and Design undergraduate minor, UCSB *Summer 2025*
Co-lecturer

Object Oriented Programming – MIAGE BSc/MSc program, Polytech Nice *2021-2023*
Teaching Assistant

Computer Graphics Introduction – MSc, Technical University of Denmark *Fall 2019*
Teaching Assistant

Service & Volunteering

Jury member *2025 - 2026*
French Informatics Society (SIF) PhD Dissertation Award

Papers program committee member *2025 - now*
SCF 2026, **Eurographics** 2026, ACM **TEI** 2025 Papers track, ACM **DIS** 2025 Pictorials track

External reviewer *2021 to now*
ACM **SIGGRAPH/SIGGRAPH ASIA** 2025, ACM **Transactions on Graphics** 2025, **Eurographics** 2024 / 2025, **Pacific Graphics** 2023 / 2024, ACM **CHI** 2024 / 2025 / 2026 (3 Special Recognitions), ACM **UIST** 2021 / 2023 / 2024 (3 Special Recognitions), ACM **DIS** 2022 / 2026 (1 Special Recognition), ACM **SCF** 2025, ACM **C&C** 2021, ACM **IMX** 2024, IEEE **ISMAR** 2024, **IEEE VR** 2025, ACM **SIGGRAPH** Posters 2023 / 2024 / 2025.

SIGGRAPH Birds of a Feather – Sustainable Graphics Research *2025*
Main organizer and discussion leader (2025), and co-organizer (2026).

Executive Committee – WiGRAPH (ACM Community Group for Women in Computer Graphics Research) *2024-now*
Online Presence and Outreach Coordinator.

Volunteer for WiGRAPH *2022-2024*
Assist in the organization of the yearly Berthouzoz Women in Research Lunch at Siggraph 2022, 2023 and 2024.

Student volunteer at conferences *2019-2022*
Eurographics 2019, JFIG 2021, Eurographics 2022

Invited talks

Designing digital representations for digital art & computer-aided manufacturing

HCI Sorbonne group seminar – *invited by Baptiste Caramiaux* — *Virtual – Oct. 2025*
LIX group seminar – *invited by Damien Rohmer and Vicky Kalogeiton* — *Saclay – Jul. 2025*
SIGCHI Paris Chapter – *invited by Theophanis Tsandilas* — *Saclay – Jul. 2025*
LIRIS HCI & Graphics research group seminar – *invited by Aurélien Tabard and Franck Davoine* — *Lyon – Jun. 2025*

Designing tools for 3D content authoring based on 3D sketching

Congrès de la SIF – *PhD award ceremony* — *Virtual – Jun. 2025*
Journées françaises de l'informatique graphique – *PhD award ceremony* — *Virtual – Oct. 2024*
Graphics And Mixed Environment Symposium webinar – *invited by Zeyu Wang* — *Virtual – Jul. 2024*
University of California Santa Barbara – *invited by Jennifer Jacobs and Tobias Höllerer* — *Santa Barbara (USA) – Aug. 2023*
University of British Columbia – *invited by Alla Sheffer* — *Vancouver (Canada) – Jul. 2023*

VideoDoodles: Hand-Drawn Animations on Videos with Scene-Aware Canvases

Inria ex)situ lab – *invited by Theophanis Tsandilas and Wendy Mackay* — *Palaiseau (France) – May 2023*
Inria MFX lab – *invited by Sylvain Lefebvre* — *Nancy (France) – May 2023*
Technical University of Denmark Christmas Workshop 2022 – *invited by J. Andreas Bærentzen* — *Lyngby (Denmark) – Dec 2022*

VR Sketching for Surface Modeling

University of Surrey SketchX virtual seminar – *invited by Yulia Gryadistskaya* — *Virtual – Sep. 2022*

CASSIE: Curve And Surface Sketching in Immersive Environments

Toronto Geometry Colloquium – *opener talk* — *Virtual – Oct. 2021*
JFIG 2020 – *presentation to the french computer graphics community* — *Virtual – Nov. 2020*

Workshops, Courses

Programming machine control systems workshop — *2025*

Hosted by Neil Gershenfeld in the "How To Make Almost Anything" class, in the MIT Media Lab
Ilan Moyer, Emilie Yu, Devon Frost, and Jennifer Jacobs

Computational craft: Computational fabrication methods for enabling craft production in textiles and ceramics — *2025*

ACM SIGGRAPH 2025 Courses
Jennifer Jacobs, Emilie Yu, Mackenzie Leake, Hannah Twigg-Smith, Nadya Peek, and Emily Whiting